ABAQUS FEA Simulation Report

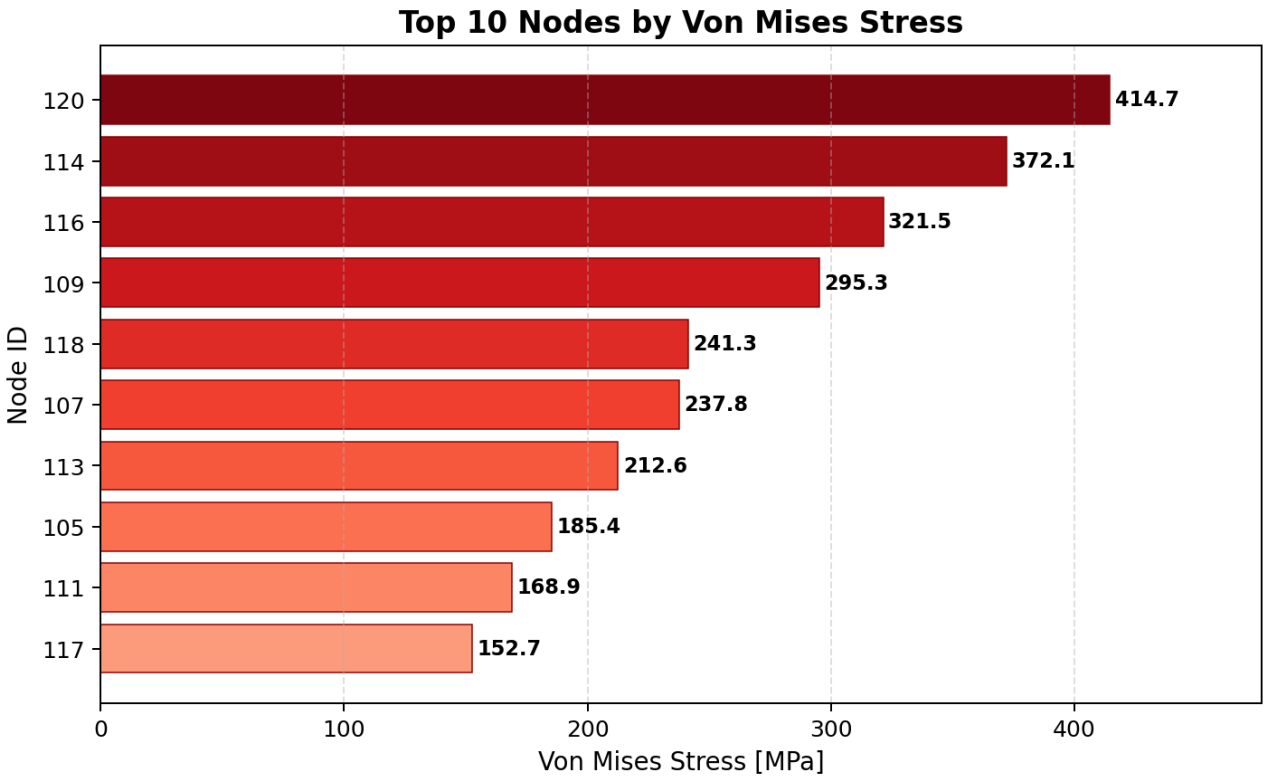
Generated: 2026-04-01 16:51
Data source: test_results.csv | Nodes analysed: 20

Critical node: 120 — Peak von Mises stress: **414.70 MPa**

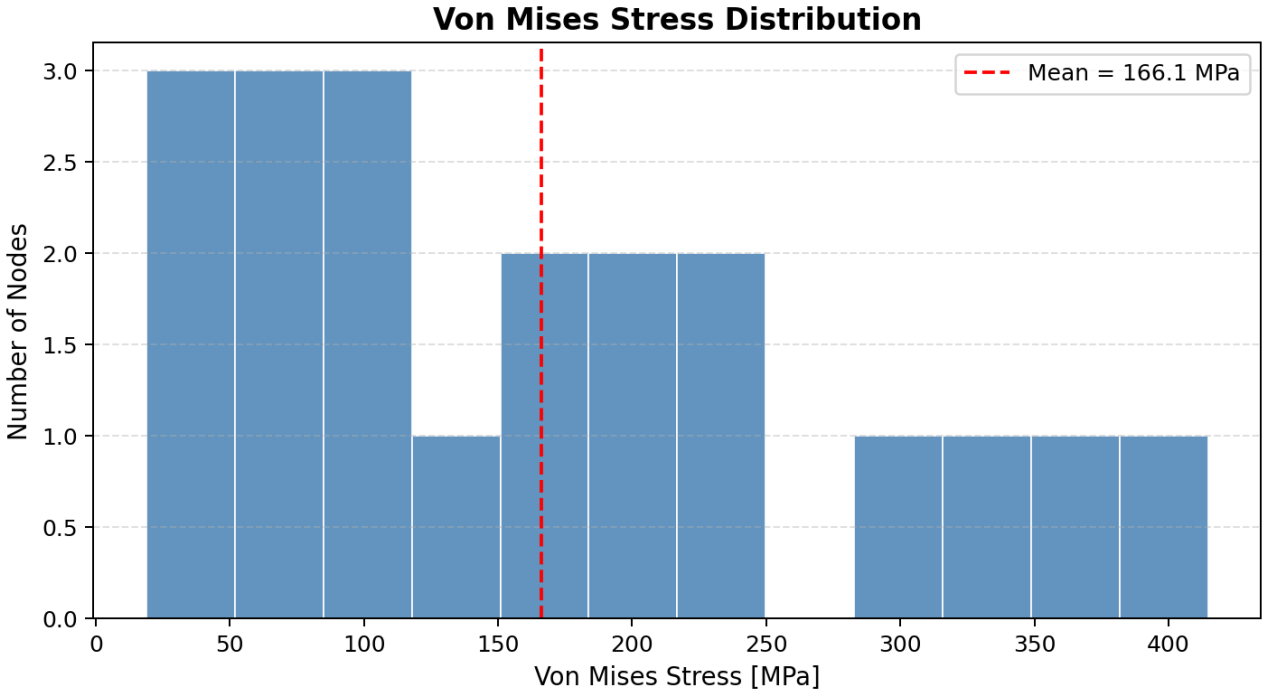
1. Stress Summary Statistics

Statistic	Value [MPa]
Minimum [MPa]	18.90
Maximum [MPa]	414.70
Mean [MPa]	166.10
Median [MPa]	143.45
Std. Deviation [MPa]	116.95

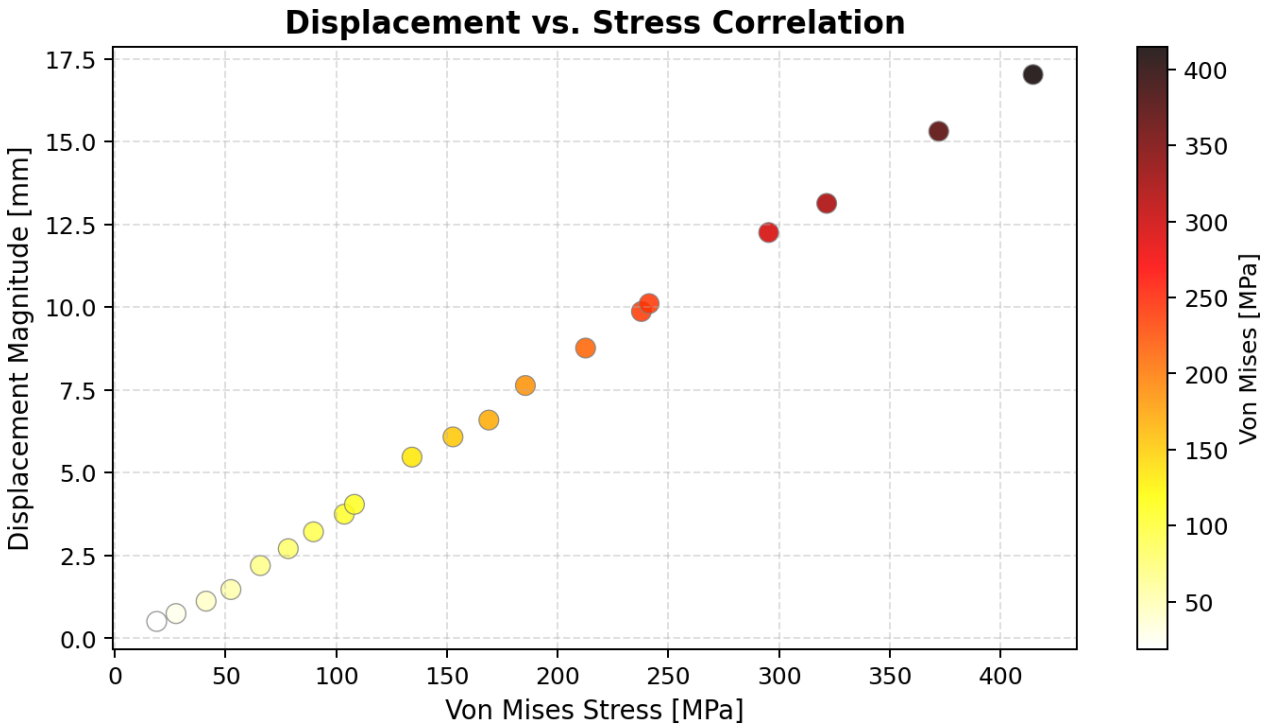
2. Top 10 Nodes by Von Mises Stress



3. Stress Distribution



4. Displacement vs. Stress Correlation



5. Complete Node Data

Node	S11	S22	S33	Mises	U1	U2	U3
101	45.2	32.1	18.7	52.4	0.0012	0.0008	0.0003
102	78.5	56.3	41.2	89.7	0.0025	0.0019	0.0007
103	112.3	87.6	63.4	134.2	0.0041	0.0033	0.0015
104	23.1	15.8	9.4	27.6	0.0006	0.0004	0.0002
105	156.7	120.3	88.1	185.4	0.0058	0.0045	0.0021
106	67.4	48.9	35.6	78.3	0.0021	0.0016	0.0006
107	198.2	152.7	112.5	237.8	0.0074	0.0059	0.0028
108	34.8	22.5	14.1	41.2	0.0009	0.0006	0.0003
109	245.6	189.4	140.2	295.3	0.0092	0.0073	0.0035
110	89.1	65.2	47.8	103.6	0.0029	0.0022	0.0009
111	142.5	108.7	79.3	168.9	0.0050	0.0039	0.0018
112	56.3	40.1	28.9	65.7	0.0017	0.0013	0.0005
113	178.4	136.8	100.5	212.6	0.0066	0.0052	0.0025
114	310.2	238.5	176.3	372.1	0.0115	0.0091	0.0044
115	15.6	10.2	6.8	18.9	0.0004	0.0003	0.0001
116	267.8	205.4	151.7	321.5	0.0099	0.0078	0.0037
117	128.9	98.3	71.6	152.7	0.0046	0.0036	0.0017
118	201.5	154.8	114.2	241.3	0.0076	0.0060	0.0029
119	92.7	68.4	49.5	108.2	0.0031	0.0024	0.0010
120	345.6	265.8	196.4	414.7	0.0128	0.0101	0.0049